Capstone First Review

Project Title	**:** Privacy-Preserving Federated Learning for Missing Person Identification
Domain	**:** Computer Vision, Federated Learning & Biometric Privacy
Reg. No	**: 1.** 23BCE9603 **2.** 23BCE9585 **3.** 23BCE9746 **4.** 23BCE8540
Project Guide	**:** Kanhaiya Sharma (**70880**)
Academic Year	**:** Fall Semester 2026–2027

1. **Final Project Title:**
   Project Title: Privacy-Preserving Federated Learning for Missing Person Identification

2. **Justification for the Selected Title:**
   The title accurately reflects the project's three core pillars:
   - **Missing Person Identification (The Application):** Addresses the critical real-world task of locating individuals across surveillance networks.
   - **Federated System (The Methodology):** Highlights the decentralized approach of training deep learning models (like YOLO) directly on edge devices, eliminating the need to centralize raw video data.
   - **Privacy-Preserving (The Security):** Emphasizes the use of advanced cryptographic techniques to secure sensitive facial biometrics against data leaks and inference attacks.

3. **Project Objectives:**
   To develop a decentralized, edge-based AI framework for missing person identification:
   Implement advanced computer vision models (such as YOLO for detection and deep neural networks for facial embeddings) across distributed surveillance networks using Federated Learning, thereby eliminating the need to centralize raw, privacy-sensitive video feeds.
   To secure sensitive biometric data against inference and reconstruction attacks:
   Integrate robust privacy-preserving techniques—such as Secure Multi-Party Computation (SMPC) and Differential Privacy (DP)—to ensure that both the facial embeddings and the aggregated model updates remain strictly confidential during transit and training.

4. **Work Distribution:**
   To ensure efficient execution and successful completion of the project, the workload has been divided among the four team members based on the core technical modules of the system.
   Member 1(G N Lokesh): Computer Vision
   - **Primary Responsibility:** Focuses on the core machine learning and image processing tasks.
   - **Specific Tasks:**
     - Pre-process and augment the surveillance/facial datasets (e.g., simulating CCTV conditions like low lighting and occlusion).
     - Implement and fine-tune the YOLO model for accurate human detection.
     - Integrate deep facial embedding models (e.g., ArcFace or FaceNet) for feature extraction.
     - Evaluate baseline recognition accuracy and false-positive rates.

   Member 2(R Kumaresh): Federated Learning
   - **Primary Responsibility:** Manages the decentralized architecture and edge-server communication.
   - **Specific Tasks:**
     - Set up the Federated Learning framework (e.g., using the Flower framework).
     - Configure the local training loops on simulated edge devices (client nodes).
     - Implement the aggregation algorithms (e.g., FedAvg or FedProx) on the central server.
     - Manage the communication pipeline between the edge nodes and the global server.

   Member 3(K Kishore): Cryptography & Privacy
   - **Primary Responsibility:** Ensures the security and confidentiality of the biometric data and model parameters.
   - **Specific Tasks:**
     - Implement Partial Secure Multi-Party Computation (P-SMPC) or Homomorphic Encryption to secure model weights before they leave the edge devices.

- Integrate Local Differential Privacy (LDP) by calibrating and injecting noise into the facial embeddings to prevent membership inference and inversion attacks.
- Evaluate and document the privacy-utility trade-off (measuring how privacy techniques impact model accuracy and latency).

Member 4(Aswin Maheswaran): UI Dashboard & Integration

- **Primary Responsibility:** Handles the user interface, database management, and end-to-end system testing.
- **Specific Tasks:**
  - Develop a user-friendly Dashboard/GUI to monitor the federated network, manage missing person alerts, and display real-time analytics.
  - Integrate the individual modules (Vision, FL, and Privacy) into a single, cohesive pipeline.
  - Conduct system-wide performance testing (e.g., calculating processing delays, communication overhead, and Equal Error Rates).
  - Lead the documentation process, including compiling the final report and formatting the literature survey.

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5. Project Timeline:

Week	Planned Activities & Tasks	Target Milestones
Week 1	• Finalize project title, problem statement, and two core objectives. • Establish team roles and distribute work among the four members.	Project Initiation
Week 2	• Complete comprehensive literature survey and identify research gaps from 15+ papers. • Draft the high-level software system architecture.	Literature Survey Completion
Week 3	• Finalize datasets (CelebA, LFW, Custom CCTV synthetic). • Prepare First Review documentation, architecture diagrams, and presentation.	**Milestone 1:** First Review & Proposal Approval
Week 4	• Develop and evaluate the baseline YOLOv8 model for human/face detection. • Implement deep facial embedding extraction using ArcFace/FaceNet.	Baseline Vision Model Ready
Week 5	• Configure the simulated edge-device software environment (client nodes). • Integrate and initialize the Federated Learning framework (Flower/flwr).	FL Environment Setup
Week 6	• Transition the local vision models (YOLO + ArcFace) into the distributed federated network. • Implement the aggregation algorithm (FedAvg/FedProx) on the central simulated server.	**Milestone 2:** Successful Decentralized Training
Week 7	• Apply Local Differential Privacy (LDP) via Opacus to noise-calibrate the facial embeddings. • Begin drafting the cryptographic layer logic (TenSEAL) for model parameters.	Privacy Module Initiation
Week 8	• Implement Partial SMPC / Homomorphic Encryption to encrypt gradient updates before transmission. • Merge Vision, FL, and Privacy modules into a single unified software pipeline.	**Milestone 3:** End-to-End System Integration
Week 9	• Prepare documentation and presentation for the Second Review. • Demonstrate the fully integrated privacy-preserving Federated Learning model.	**Milestone 4:** Second Review Completion
Week 10	• Develop the interactive Streamlit Dashboard/GUI for query dispatch and	Performance

	binary alerts. • Evaluate software system metrics (Accuracy, EER) and optimize the privacy-utility trade-off.	Optimization
Week 11	• Finalize GUI and conduct rigorous end-to-end software system testing. • Compile test results, generate charts, and draft the final project report.	Final Testing & Draft Ready
Week 12	• Finalize the project report and presentation slides. • Conduct the final project demonstration.	**Milestone 5:** Final Review & Submission

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6. Hardware and Software Requirements:

Hardware Requirements:

- **Primary Compute Platform (Cloud):** AWS EC2 g4dn.xlarge instance (4 vCPUs or higher, 16 GB RAM or higher, 1x NVIDIA T4 16GB GPU or higher).
- **Cloud Storage:** AWS Elastic Block Store (EBS gp3) with 100 GB of storage or higher (scalable to 500 GB or higher).
- **Local Development Machine:** Intel Core i5 / AMD Ryzen 5 processor or higher, 8 GB RAM or higher, and 256 GB SSD or higher (utilized strictly for code development and Remote SSH access).

Software Requirements:

- **Operating System (Cloud):** Ubuntu 20.04 LTS or higher.
- **Ubuntu OS Minimum Requirements:** 2 GHz dual-core processor or higher, 4 GB RAM or higher (8 GB RAM or higher recommended for development environments), and 25 GB of free disk space or higher.
- **Operating System (Local):** Windows 10/11 or higher, or macOS 10.15 or higher.
- **Python:** Version 3.8 or higher.
- **Deep Learning:** PyTorch (torch 1.13.0 or higher, torchvision 0.14.0 or higher).
- **Computer Vision:**
  - Ultralytics YOLO (ultralytics 8.0.0 or higher)
  - InsightFace (insightface 0.7.0 or higher)
  - OpenCV (opencv-python 4.5.0 or higher)
- **Federated Learning:** Flower Framework (flwr 1.0.0 or higher).
- **Cryptography and Privacy:**
  - Opacus (opacus 1.2.0 or higher for Differential Privacy)
  - TenSEAL (tenseal 0.3.0 or higher for Homomorphic Encryption / SMPC)
- **Dashboard/GUI:** Streamlit (streamlit 1.10.0 or higher).
- **IDE:** Visual Studio Code (v1.70 or higher) with the "Remote - SSH" extension.
- **Version Control:** Git (v2.20 or higher) and GitHub.
- **Environment Management:** Miniconda / Conda (v4.12.0 or higher).

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7. Dataset Information:

To train, evaluate, and test the proposed Privacy-Preserving Federated AI System, the project utilizes a combination of large-scale standard biometric datasets and a curated custom surveillance dataset.

7.1. CelebFaces Attributes (CelebA) Dataset:

- **Dataset Name and Source:** CelebFaces Attributes (CelebA) Dataset, provided by MMLAB, The Chinese University of Hong Kong (CUHK).
- **Dataset Size:** Approximately 1.63 GB.
- **Number of Samples/Instances:** 202,599 celebrity images across 10,177 unique identities.
- **Features/Attributes:** Each image is annotated with 40 binary attributes (e.g., Bald, Eyeglasses, Smiling), 5 facial landmark locations, and identity markers.
- **Data Format:** JPEG images with a resolution of 178x218 pixels.
- **Training and Testing Data:**
  - Standard partition includes ~162,770 training images, ~19,867 validation images, and ~19,962 test images.

    - *Project-Specific Usage:* As per the federated learning methodology, subsets (e.g., 500 to 1,000 identities) will be partitioned with a 90% training and 10% testing split distributed across simulated edge devices.
- **Dataset Source/Link:** CelebA Official Link

7.2. Labeled Faces in the Wild (LFW) Dataset:

- **Dataset Name and Source:** Labeled Faces in the Wild (LFW), provided by the University of Massachusetts, Amherst.
- **Dataset Size:** Approximately 180 MB.
- **Number of Samples/Instances:** 13,233 unconstrained face images of 5,749 people.
- **Features/Attributes:** Identity labels designed specifically for evaluating unconstrained face verification and recognition in real-world scenarios.
- **Data Format:** 250x250 JPEG images.
- **Training and Testing Data:** Evaluated using 10-fold cross-validation or specific subsets (e.g., isolating 5 training and 5 testing images per identity for face verification testing).
- **Dataset Source/Link:** LFW Official Link | Kaggle

7.3. Composite Synthetic Surveillance Dataset (Custom):

- **Dataset Name and Source:** Custom Simulated CCTV Dataset, curated by combining publicly available in-the-wild datasets (CrowdHuman, WiderPerson, and CelebA).
- **Number of Samples/Instances:** 780 composite group images (240 positive matches, 540 negative decoys).
- **Features/Attributes:** Designed to realistically emulate CCTV conditions, incorporating low illumination, occlusions, and clothing-matched decoys. Each target identity is restricted to only 3-4 reference images to maintain privacy constraints.
- **Data Format:** Annotated bounding box data (compatible with Ultralytics YOLOv8-v12).
- **Training and Testing Data:** Utilized primarily to test baseline YOLO detection performance and the cosine similarity of ArcFace/FaceNet embeddings under surveillance distortions.